Health, Just a Tap Away!

The MediCare App

- The MediCare App is a digital healthcare platform designed to connect patients with doctors for both online consultations and offline clinic visits.
- The application data is managed using SQL in a relational database environment.
- The system consists of two relational tables Doctors and Consultations.
- Doctor information is stored in a master doctors table, while consultation details are maintained in a separate consultations table to ensure data normalization and avoid redundancy.

How does this System works?

- The Doctors table includes doctor_id as the primary key, along with doctor_name, specialization, city, and available_mode indicating whether the doctor is available online or offline.
- The Consultations table includes consultation_id as the primary key, along with patient_name, age, gender, consultation_date, consultation_type indicating online or offline mode, doctor_id as a foreign key, and fees.
- The doctor_id acts as the primary key in the Doctors table and as a foreign key in the Consultations table.

Problem Statement

- Retrieve consultation records where the consultation fee exceeds 1,000.
- Identify doctors who are exclusively available for online consultations.
- Retrieve the 10 most recent consultation records based on consultation date.
- Compare total revenue generated from online and offline consultations.

Retrieve consultation records where the consultation fee exceeds 1,000.

SELECT*FROM consultations WHERE fees > 1500;

consultation_id	patient_name	age	gender	consultation_date	consultation_type	doctor_id	fees
5	Neha Joshi	28	Female	2024-01-09	Online	5	1800
16	Sunil Pawar	50	Male	2024-01-20	Online	5	1800
24	Rahul Singh	35	Male	2024-01-28	Online	13	2000
26	Vivek Mishra	46	Male	2024-02-01	Online	15	1600
30	Ajay Verma	52	Male	2024-02-05	Online	19	2000
33	Pooja Patil	30	Female	2024-02-08	Online	22	1600
37	Komal Shah	32	Female	2024-02-12	Online	26	2000
41	Riya Sharma	25	Female	2024-02-16	Online	30	1600
42	Aman Verma	40	Male	2024-02-17	Offline	30	1600
43	Pooja Patil	30	Female	2024-02-18	Online	22	1600
44	Rahul Singh	35	Male	2024-02-19	Offline	15	1600
46	Rakesh Yadav	58	Male	2024-02-21	Offline	5	1800
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Identify doctors who are exclusively available for online consultations.

SELECT doctor_name, specialization, city, available_mode FROM doctors WHERE available_mode = 'Online';

doctor_name	specialization	city	available_mode
Dr. Amit Shah	Cardiologist	Mumbai	Online
Dr. Rahul Mehta	Orthopedic	Pune	Online
Dr. Arjun Rao	Neurologist	Bangalore	Online
Dr. Kunal Joshi	Cardiologist	Ahmedabad	Online
Dr. Ritu Malhotra	Dermatologist	Delhi	Online
Dr. Anjali Patil	Gynecologist	Pune	Online
Dr. Kavita Nair	Pediatrician	Kochi	Online
Dr. Ayesha Shaikh	Dermatologist	Mumbai	Online
Dr. Nidhi Agarwal	General Physician	Jaipur	Online
Dr. Meenal Joshi	Gynecologist	Indore	Online
Dr. Rohan Malviya	Pediatrician	Bhopal	Online
Dr. Mohit Bansal	Cardiologist	Delhi	Online
Dr. Akash Verma	General Physician	Lucknow	Online
Dr. Nitin Chawla	Neurologist	Gurgaon	Online
Dr. Harish Menon	Pediatrician	Trivandrum	Online
Dr. Anoop Mishra	Cardiologist	Kanpur	Online

Retrieve the 10 most recent consultation records based on consultation date.

SELECT *FROM consultations ORDER BY consultation_date DESC
LIMIT 10;

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content

consultation_id	patient_name	age	gender	consultation_date	consultation_type	doctor_id	fees
50	Alok Mishra	51	Male	2024-02-25	Offline	1	800
49	Nikita Deshmukh	29	Female	2024-02-24	Online	9	600
48	Manoj Gupta	47	Male	2024-02-23	Offline	3	900
47	Shweta Kulkarni	34	Female	2024-02-22	Online	12	900
46	Rakesh Yadav	58	Male	2024-02-21	Offline	5	1800
45	Neha Joshi	28	Female	2024-02-20	Online	7	1000
44	Rahul Singh	35	Male	2024-02-19	Offline	15	1600
43	Pooja Patil	30	Female	2024-02-18	Online	22	1600
42	Aman Verma	40	Male	2024-02-17	Offline	30	1600
41	Riya Sharma	25	Female	2024-02-16	Online	30	1600
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Compare total revenue generated from online and offline consultations.

```
SELECT  d.available_mode,  SUM(c.fees) AS total_revenue FROM
consultations c INNER JOIN doctors  d  ON c.doctor_id = d.doctor_id
GROUP BY d.available_mode;
```

	available_mode	total_revenue
▶	Online	35100
	Offline	25100

Conclusion

- The MediCare App provides an efficient platform for online and offline doctor consultations.
- The relational database design ensures organized data storage and avoids redundancy. SQL queries help generate meaningful insights for decision-making and performance analysis.
- Overall, the system improves healthcare accessibility, efficiency, and data management.

Thankyou